

a) How many classes are there in the code?

There are 4 classes, Queue, Checker, Store, and SimApp.

b) What is the meaning of each class?

- Queue class: It handles the insertion, removal, and display of the queue.
- Checker class: Represent a cashier and it uses the queue class to manage its lines of customers.
- Store class: This class manages multiple checker objects and handles assigning customers to the shortest lines to ensure efficient queue management.
- SimApp class: It is the main class and allows the user to choose from a menu of items.

c) Describe the meaning of the program.

This program simulates a supermarket checkout system where multiple checkout lines (represented by Checker objects) manage customer queues. The user can add new customers, advance time, and display the current state of all checkout lines.

d) What can this program be used for?

- Simulate customer flow in a retail store.
- Study and analyze wait times, line balancing, and service efficiency.